Report Writing and Implementation Paper Writing

VTU Guidelines

The project report has to be organized in the following order.

1.Cover Page / Title Page (Includes your college name, VTU emblem, project title, student names, USN, guide name, and HOD name).

2. Inside Title Page

3. Certificate (Signed by your guide, HOD, and principal).

4. Declaration signed by the Candidate

5.Acknowledgement (Brief appreciation to your guide, HOD, and college staff).

6.Abstract (A comprehensive summary of the project in about 300–500 words).

7.Table of Contents (With page numbers for chapters, sections, and subsections).

8.List of Tables

9. List of Figures (Including abbreviations and nomenclatures, if any)

10. Abbreviations/ Notations/ Nomenclature (if any)

11. Text of the Report

Chapter 1

Chapter 2

…..

…..

12. References

13. Appendices (if any)

14. Non-paper materials (if any) (CDs etc)

Core Chapters Breakdown

Chapters should be logically divided to detail your project from start to finish

Chapter 1: Introduction -

Project background, problem statement, objectives, and the overall scope.

Chapter 2: Literature Survey -

Review of existing research papers, journals, or related systems (minimum 10-12 references).

Table format:

Standard Format

S.No	Author(s) & Year	Title / Focus	Method / Technique Used	Dataset	Key Results	Limitations / Research Gap
1						
2						

Chapter 3: System Requirements & Design-

Software requirement specification (SRS) and hardware requirements.

System architecture, block diagrams, Data Flow Diagrams (DFDs), UML diagrams.

Chapter 4: Implementation-

Details of the technologies and programming languages used.

Division of modules and brief descriptions.

Chapter 5: Testing and Results / Implementation Details

Testing procedures and test cases.
Output analysis, code snippets, and relevant screenshots (for software) or circuit/prototype diagrams (for hardware).

Chapter 6: Conclusion and Future Scope

Summary of project achievements, limitations, and future enhancements.

References / Bibliography

All citations must be strictly formatted in **IEEE** or **APA** style.

IEEE style eg:

[3] S. P. Mohanty, D. P. Hughes, and M. Salathé, "Using deep learning for image-based plant disease detection," Frontiers in Plant Science, vol. 7, p. 1419, 2016.

If reference is website…give full URL as follows:

[12] TensorFlow, "TensorFlow Lite for mobile and edge devices," TensorFlow. [Online]. Available: https://www.tensorflow.org/lite. [Accessed: 10-Jul-2026].

[13] A. Rosebrock, "Deep learning for image classification with Keras," PyImageSearch. [Online]. Available: https://pyimagesearch.com/deep-learning-image-classification/. [Accessed: 12-Jul-2026].

If no individual author exists (common for datasets/orgs):

[14] Kaggle, "PlantVillage Dataset," Kaggle. [Online]. Available: https://www.kaggle.com/datasets/plantvillage. [Accessed: 05-Jul-2026].

Appendix

Contains lengthy source code, large data tables, and data sheets

Details of each Section

1. Problem Statement

A **problem statement** is a concise declaration of the issue or challenge that initiates the research or design activity. It should clearly articulate the motivation and background of the project. When preparing a problem statement, students should address:

a. **What** is the issue to be addressed (problem or question)?
b. **Why** is it important to address this issue?
c. **How** can the proposed project provide a solution?
d. **Who** will benefit from the project outcomes?

Clearly defining the scope ensures the project remains **feasible, realistic, and achievable** within the allotted resources and timeline.

2. Literature Review

The literature review is a structured discussion of published information relevant to a specific subject area. Its primary purpose is to **summarize**, **synthesize**, and **critically analyze** the ideas and findings of others, establishing the foundation for the student's own project work.

Purpose

- To understand existing work in the selected domain.
- To identify **gaps**, **limitations**, and **opportunities** for further study.
- To position the student's project in the context of prior research and design.

The scope delineates the boundaries and constraints of the project to ensure clarity and feasibility. Scope may include:

- Timeframe: Duration of the study or development cycle.
- Geographical limits: If applicable, specify the location or context.
- Environmental conditions: Operational or testing environments.
- Functional boundaries: Specific features, modules, or systems under study.

Purpose of defining scope:

- Establish a shared understanding among stakeholders (students, guides, evaluators).
- Ensure the project remains achievable, focused, and realistic.

Methodology

The **Methodology** section defines the technical approach adopted to design and execute the research component of the project. It focuses on experimental, analytical, and computational procedures, excluding administrative or non-technical activities such as proposal submission or report writing.

When preparing the methodology, students should address the following:

1. **Objective of the Study**
    - Clearly state the aim of the investigation.
    - Example: *Given a new design idea, evaluate its performance in terms of sensitivity, accuracy, processing time, etc.*
2. **Measurement Parameters**
    - Specify the parameters to be measured (e.g., time, storage size, current, cost, sensitivity, accuracy).
3. **Measurement Method**
    - Describe the method of measurement.
    - Example: *Build a prototype and measure sensitivity directly, or indirectly using an equation/method suggested in published literature.*

4. **Tools and Equipment**
    - List simulation software, instruments, prototypes, or other resources required for the experiment.
5. **Experimental Procedure**
    - Describe step-by-step how measurements will be taken and recorded.
    - Include safety precautions, calibration methods, and repetition procedures if applicable.
6. **Error Sources and Mitigation**
    - Identify potential sources of error or conditions that could interfere with the measurement.
    - Suggest measures to minimize these errors.
7. **Data Analysis**
    - State how the collected data will be processed and interpreted.
    - Specify statistical tools, calculations, graphs, tables, or figures to be used.

The methodology must be **precise, replicable, and technically detailed**, allowing another researcher to repeat the study and obtain similar results.

Result, Analysis, and Discussion

Upon completing the experimental procedure as outlined in the Methodology section, the results should be compiled clearly and concisely, typically using **tables**, **graphs**, or **figures** to summarize the measurements and observations.

The **Analysis and Discussion** stage involves interpreting and evaluating these results. Suggested discussion points include:

1. **Trend Observation**
    - Describe how the variable(s) of interest change in relation to other variables.
    - Indicate whether these trends align with theoretical expectations or prior findings.
2. **Academic Interpretation**
    - Provide an evidence-based interpretation of results, supported by relevant theory,

literature comparisons, or logical reasoning.

- Highlight agreements, deviations, or improvements over existing work.

3. **Significance and Implications**
   - Discuss the importance of the findings in the academic or practical context.
   - Explain the broader impact on the field, industry, or society.
4. **Potential Applications**
   - Identify areas where the findings could be applied.
   - Mention any adaptations needed for real-world implementation.

This section must focus on **technical and analytical aspects**. Non-technical activities such as report formatting, supervisor meetings, or administrative submissions should not be included.

1. **Number of Copies of Project Report**

   The total number of report to be prepared are

   - One copy to the department
   - One copy to the guide(s)
   - Two copies to the sponsoring agency if any
   - One copy to the candidate

2. **Page Setup**

   Project reports should be typed neatly only on one side of the paper with

   - 1.5 or double line spacing on a A4 size bond paper (210 x 297 mm).
     - The margins should be:
     - Left – 1.25″,
     - Right – 1″,
     - Top and Bottom – 0.75″

3. **Approval from the guide:**
   - Before taking the final printout, the approval of the concerned guide(s) is mandatory and suggested corrections, if any, must be incorporated.
4. **For making Copies**
   - For making copies, dry toner Xerox is suggested.

5. **Contents of the Project Report**

   Every copy of the report must contain
   - Inner title page (White)
   - Outer title page with a plastic cover
   - Certificate in the format enclosed both from the college and the organization where the project is carried out.
   - An abstract (synopsis) not exceeding 100 words, indicating salient features

6. **The organization of the report should be as follows**
    - Inner title page
    - Abstract or Synopsis
    - Acknowledgments
    - Table of Contents
    - List of table & figures (optional)

All above pages usually are numbered in roman

7. **Chapters (to be numbered in Arabic)** containing Introduction-, which usually specifies the scope of work and its importance and relation to previous work and the present developments, Main body of the report divided appropriately into chapters, sections and subsections.
8. **The Last chapter :** should contain the summary of the work carried, contributions if any, their utility along with the scope for further work.

9. The chapters, sections and subsections may be numbered in the decimal form for e.g.
   - Chapter 2,
   - sections as 2.1, 2.2 etc., and
   - subsections as 2.2.3, 2.5.1 etc.

10. **Font Size**:

Sl. No.	Content	Justification	Font Size
01	The chapter must be justified	Left to Right	16
02	Followed by the title chapter	Centered	18
03	Section/Subsection numbered along with their headings	Left	16
04	Subsection and Heading	Left	14
05	The body or text of the report	Justified	12

11. **Numbering of the figures and Table:**

The figures and tables must be numbered chapter wise for e.g. Fig.2.1 Block diagram of a serial binary adder, Table 3.1 Primitive flow table etc

serial binary adder, Table 3.1 Primitive flow table etc

11. Reference or Bibliography:

The references should be numbered serially in the order of their occurrence in the text and their numbers should be indicated within square brackets for e.g. [3]. The section on references should list them in serial order in the following format.

For textbooks – A.V. Oppenheim and R.W. Schafer, Digital Signal Processing, Englewood, N.J., Prentice Hall, 3 Edition, 1975.

For papers – Devid, Insulation design to combat pollution problem, Proc of IEEE, PAS, Vol 71, Aug 1981, pp 1901-1907.

12. Equations and units

Only SI units are to be used in the report. Important equations must be numbered in decimal form for e.g.

$$V = IZ \ldots\ldots\ldots \quad (3.2)$$

All equation numbers should be right-justified.

Sample References

REFERENCES

[1] J. Friedman, T. Hastie, R. Tibshirani *et al.*, *The elements of statistical learning*. Springer series in statistics New York, 2001, vol. 1, no. 10.

[2] M. Mohri, A. Rostamizadeh, and A. Talwalkar, *Foundations of machine learning*. MIT press, 2018.

[3] J. C. Schlimmer and R. H. Granger, "Incremental learning from noisy data," *Mach. Learn.*, vol. 1, no. 3, pp. 317–354, 1986.

[4] B. Schölkopf, D. Janzing, J. Peters, E. Sgouritsa, K. Zhang, and J. Mooij, "On causal and anticausal learning," in *29th International Conference on Machine Learning (ICML 2012)*. International Machine Learning Society, 2012, pp. 1255–1262.

[5] M. Salganicoff, "Tolerating concept and sampling shift in lazy learning using prediction error context switching," *Artif. Intell. Rev.*, vol. 11, no. 1-5, pp. 133–155, 1997.

[6] A. Storkey, "When training and test sets are different: characterizing learning transfer," *Dataset shift in machine learning*, vol. 30, pp. 3–28, 2009.

[7] M. Kull and P. Flach, "Patterns of dataset shift," in *First International Workshop on Learning over Multiple Contexts (LMCE) at ECML-PKDD*, 2014.

[8] R. Elwell and R. Polikar, "Incremental learning of concept drift in nonstationary environments," *IEEE Trans. Neural Networks*, vol. 22, no. 10, pp. 1517–1531, 2011.

[9] L. M. Rose, R. F. Paige, D. S. Kolovos, and F. A. Polack, "The epsilon generation language," in *European Conference on Model Driven Architecture-Foundations and Applications*. Springer, 2008, pp. 1–16.

[10] D. S. Kolovos and R. F. Paige, "Towards a modular and flexible human-usable textual syntax for emf models." in *MODELS Workshops*, 2018, pp. 223–232.

Name & Signature of the Guide Name & Signature of the Head of organization

COLOUR OF THE OUTER COVER/FRONT PAGE OF UG DISSERTATION / PROJECT REPORT

Sl.No.	UG course	Color of the outer cover/front page of the report
1	Electronics & Communication, Telecommunication, Bio-Medical, Medical Electronics, Electrical & Electronics and Instrumentation Technology (EC/TE/BM/ML/EE/IT)	PURPLE
2	Computer Science and Information Science and Engineering (CS/IS)	CREAM
3	Mechanical, Printing Technology, Mining, Industrial Production,	SKY BLUE

IMPLEMENTATION PAPER WRITING

How to Write an IEEE Implementation Paper (Machine Learning)

An "implementation paper" reports on a system you built and tested — not just a theoretical proposal. IEEE conference/journal papers follow a fairly fixed structure (IEEEtran template, two-column format). Below is each section, what it should contain, and a worked example using a sample topic: **"Implementation of a CNN-Based Model for Real-Time Plant Disease Detection."**

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1. Title

- Concise (10–15 words), states the technique + application/domain.
- Avoid vague words like "Study on..." — be specific about what was implemented.

Example:

> "Implementation of a Lightweight CNN Model for Real-Time Plant Leaf Disease Detection Using Transfer Learning"

2. Author Names & Affiliations

- Format: Name, Department, Institution, City, Country, Email.
- IEEE uses a specific two-column header block — the template handles formatting automatically.

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3. Abstract (150–250 words)

Must summarize the **entire paper** in miniature:

1. Problem/motivation (1 sentence)
2. Gap in existing work (1 sentence)
3. What you implemented (method/model)
4. Dataset & key implementation detail
5. Results (with a number — accuracy, F1, latency, etc.)
6. One-line significance/conclusion

(Note: no citations in the abstract

4. Keywords

- 4–6 IEEE-indexed terms, comma-separated.

Example:

> *Keywords*— Convolutional Neural Network, Transfer Learning, Plant Disease Detection, Edge Deployment, MobileNetV2, TensorFlow Lite

5. I. Introduction

Structure (funnel shape — broad to narrow):

1. **Context** – why the domain matters (real-world problem, statistics)
2. **Problem statement** – what's unsolved/inefficient
3. **Motivation** – why ML/your approach helps
4. **Brief mention of related work gap** – 2–3 sentences, save details for Section II
5. **Contributions** – a clearly bulleted list (very common in implementation papers)
6. **Paper organization** – "Section II reviews related work, Section III describes the methodology..." etc.

Example (contributions bullet, typical IEEE style):

> "The key contributions of this paper are as follows:
> - We implement a transfer-learning-based CNN pipeline optimized for low-resource devices.
> - We benchmark the model against VGG16 and ResNet50 baselines on accuracy, size, and latency.
> - We deploy and test the model on a Raspberry Pi 4 under real field lighting conditions."

6. II. Related Work / Literature Review

- Organize **thematically**, not chronologically (e.g., "Traditional ML approaches," "Deep CNN approaches," "Lightweight/edge models").
- Every reference should be critically discussed — what they did, and their **limitation** that motivates your work.
- End with a short paragraph explicitly stating the gap your paper fills.

Example:

> "Mohanty et al. [3] applied AlexNet and GoogLeNet on the PlantVillage dataset, achieving 99.35% accuracy; however, their models required over 500 MB of storage, making them impractical for edge deployment. Ramcharan et al. [5] used MobileNet for cassava disease detection on smartphones but did not evaluate real-time inference latency. Unlike these works, the present study focuses on both accuracy *and* deployability, explicitly benchmarking inference time on constrained hardware."

7. III. Proposed Methodology / System Architecture

This is the **design** section — describe *what* you built before *how* you built it (implementation details come next). Include:

- System/block diagram (figure)
- Dataset description (source, size, classes, split ratio, preprocessing)
- Model architecture (layers, backbone, hyperparameters)
- Mathematical formulation (loss function, optimizer equations) if relevant

Example subsection layout:

> **A. Dataset** The PlantVillage dataset consists of 54,305 labeled leaf images across 38 disease classes. Images were resized to 224×224 and split 80/10/10 for train/validation/test.
> **B. Model Architecture** MobileNetV2 was used as the backbone (pretrained on ImageNet), with the top layer replaced by a Global Average Pooling layer, a Dense(128, ReLU) layer, and a softmax output layer for 38 classes.
>
> **C. System Architecture** Fig. 1 shows the end-to-end pipeline: image capture → preprocessing → CNN inference → TFLite conversion → deployment on Raspberry Pi.

8. IV. Implementation

This is the section unique to **implementation papers** (as opposed to purely theoretical ones). Be concrete and reproducible:

- Hardware/software environment (GPU, RAM, OS, framework + version)
- Training configuration (epochs, batch size, learning rate, optimizer)
- Tools/libraries (TensorFlow 2.x, PyTorch, scikit-learn, OpenCV, etc.)
- Any deployment pipeline (APIs, edge devices, containers)
- Pseudocode or algorithm box, if applicable

Example:

> "The model was trained using TensorFlow 2.12 on an NVIDIA RTX 3060 GPU (12 GB VRAM), with Adam optimizer (lr = 0.0001), batch size 32, over 25 epochs with early stopping (patience = 5). Data augmentation (rotation ±20°, horizontal flip, zoom 0.2) was applied using `ImageDataGenerator`. Post-training, the model was quantized to 8-bit integer precision using TensorFlow Lite Converter and deployed on a Raspberry Pi 4 (4 GB RAM) running Raspbian OS, interfaced with a Pi Camera Module v2 for real-time capture."

9. V. Results and Discussion

- Present quantitative results with **tables and graphs** (confusion matrix, accuracy/loss curves, comparison tables).
- Compare against baseline(s) — this is essential for an implementation paper.
- Discuss *why* results occurred, not just what they were.
- Include metrics standard to ML: Accuracy, Precision, Recall, F1-score, Confusion Matrix, ROC-AUC, inference time, model size.

Example table caption:

> TABLE I. Performance Comparison of Models

Model	Accuracy	Size (MB)	Inference Time (ms)
VGG16	97.1%	528	620
ResNet50	96.8%	98	340
Proposed (MobileNetV2)	**96.3%**	**12**	**180**

10. VI. Conclusion and Future Work

- Restate the problem and what was implemented (1–2 sentences).
- Summarize key results (with numbers).
- State limitations honestly.
- Suggest concrete future work (not vague "more research needed").

Example:

> "This paper presented the implementation of a lightweight CNN-based system for real-time plant disease detection, achieving 96.3% accuracy with significantly reduced computational overhead compared to standard architectures. The system was successfully deployed and tested on a Raspberry Pi 4 under field conditions. Future work will extend the model to multi-crop datasets and explore federated learning to enable on-device model updates without centralized data collection."

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11. Acknowledgment (optional)

> "This work was supported by [Grant/Institution name]."

12. References

- IEEE numbered style `[1]`, `[2]`... in order of first citation, not alphabetical.
- Format (journal): `[1] A. Author, B. Author, "Title of paper," Journal Name, vol. X, no. Y, pp. XX-XX, Month Year.`
- Format (conference): `[2] A. Author, "Title of paper," in Proc. Conf. Name, City, Country, Year, pp. XX-XX.`

Example:

> [1] S. P. Mohanty, D. P. Hughes, and M. Salathé, "Using deep learning for image-based plant disease detection," *Frontiers in Plant Science*, vol. 7, p. 1419, 2016.

REFERENCES

[1] J. Friedman, T. Hastie, R. Tibshirani *et al.*, *The elements of statistical learning*. Springer series in statistics New York, 2001, vol. 1, no. 10.

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[10] D. S. Kolovos and R. F. Paige, "Towards a modular and flexible human-usable textual syntax for emf models." in *MODELS Workshops*, 2018, pp. 223–232.

THANK YOU